

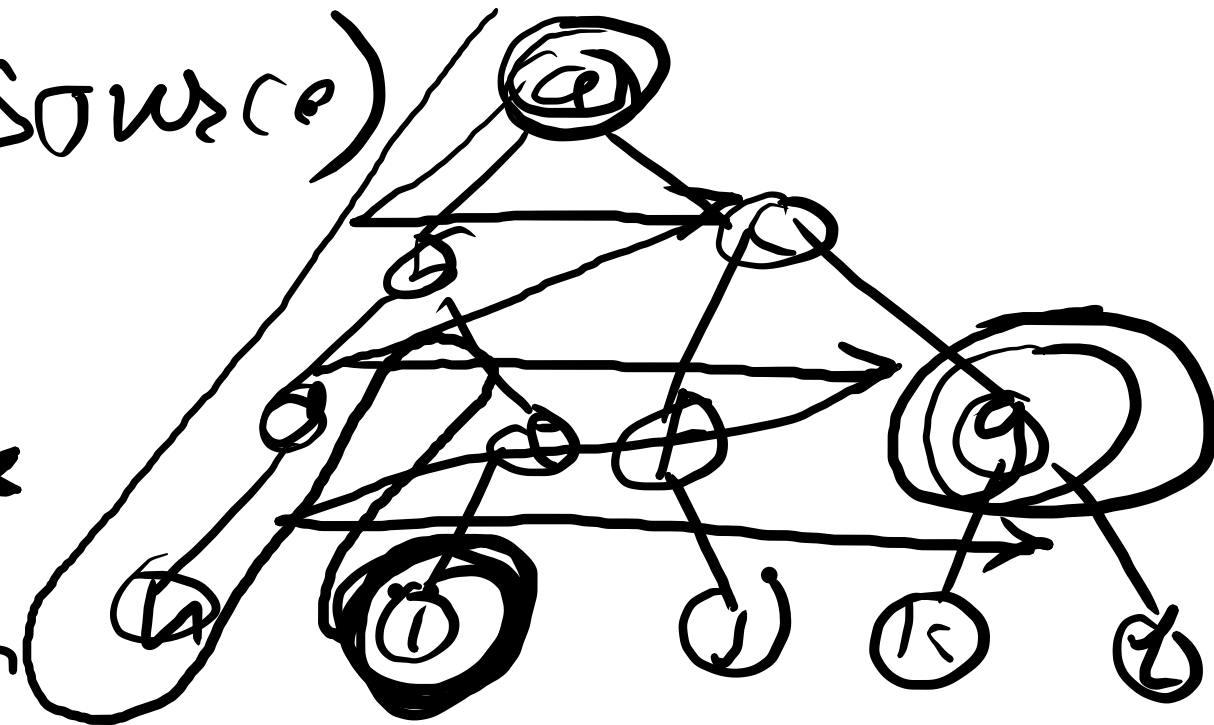
Graph Algorithm :-

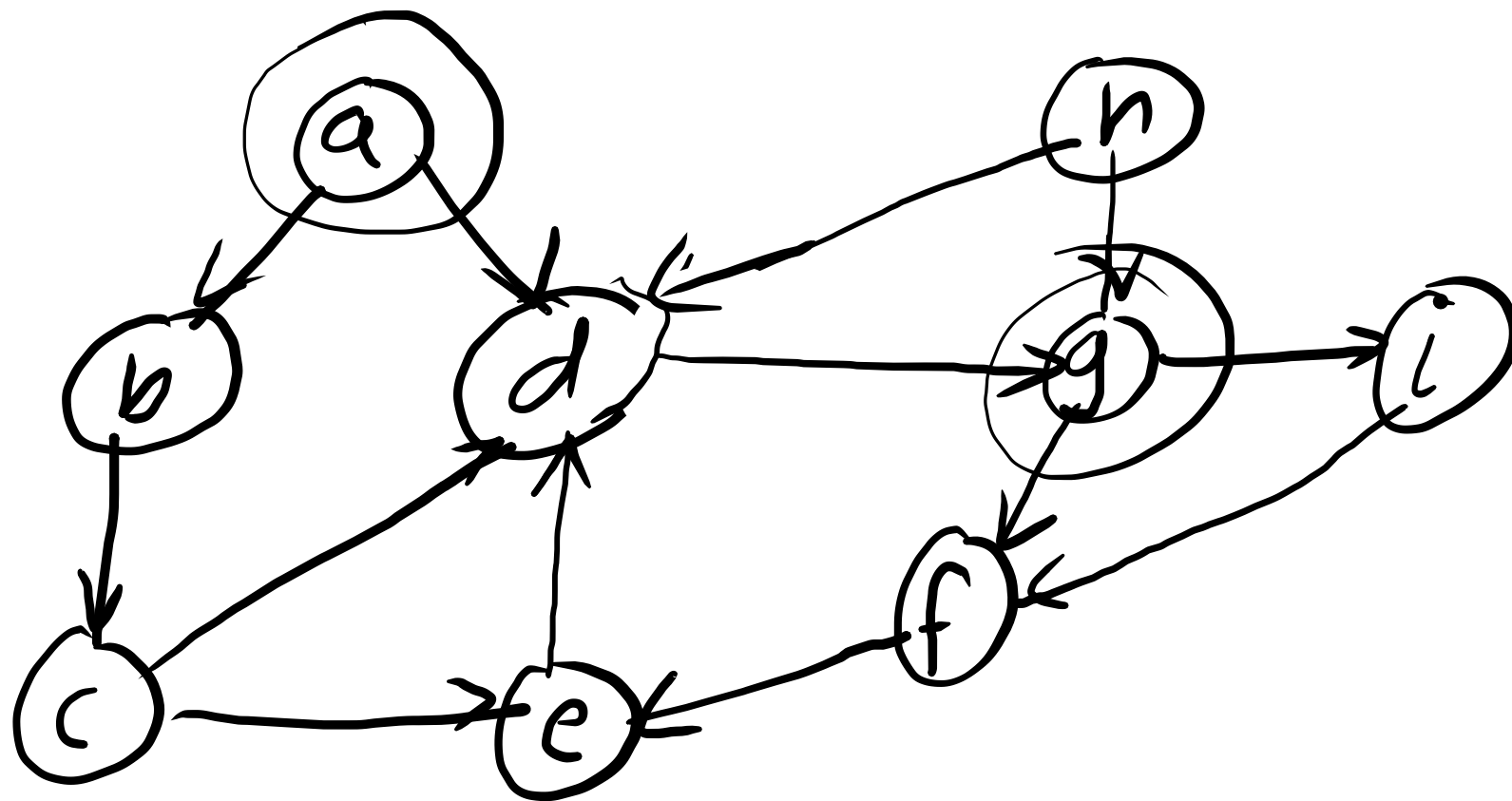
Breadth First Search :-

✓ Adjacency List
✓ Adjacency Matrix

Shortest path (Single Source)

Given two vertices $u, v \in V$
the shortest path from u to v
is a path that has minimum
length among all paths from u to v





Single Source Shortest Path
with unit weights

a to g

$a \rightarrow b \rightarrow c \rightarrow d \rightarrow g$
 length 4

$a \rightarrow b \rightarrow c \rightarrow e \rightarrow d \rightarrow g$
 length 5

$a \rightarrow d \rightarrow g$
 length 2

BFS in Graph:

- Initially color all vertices as WHITE (^{Not}_{considered})
- Create Empty BFS Tree T
- Create a Q , Insert source vertex S in Q and color S GRAY (GRAY = in Q)
- Make S root of tree T
- Repeat until Q is empty

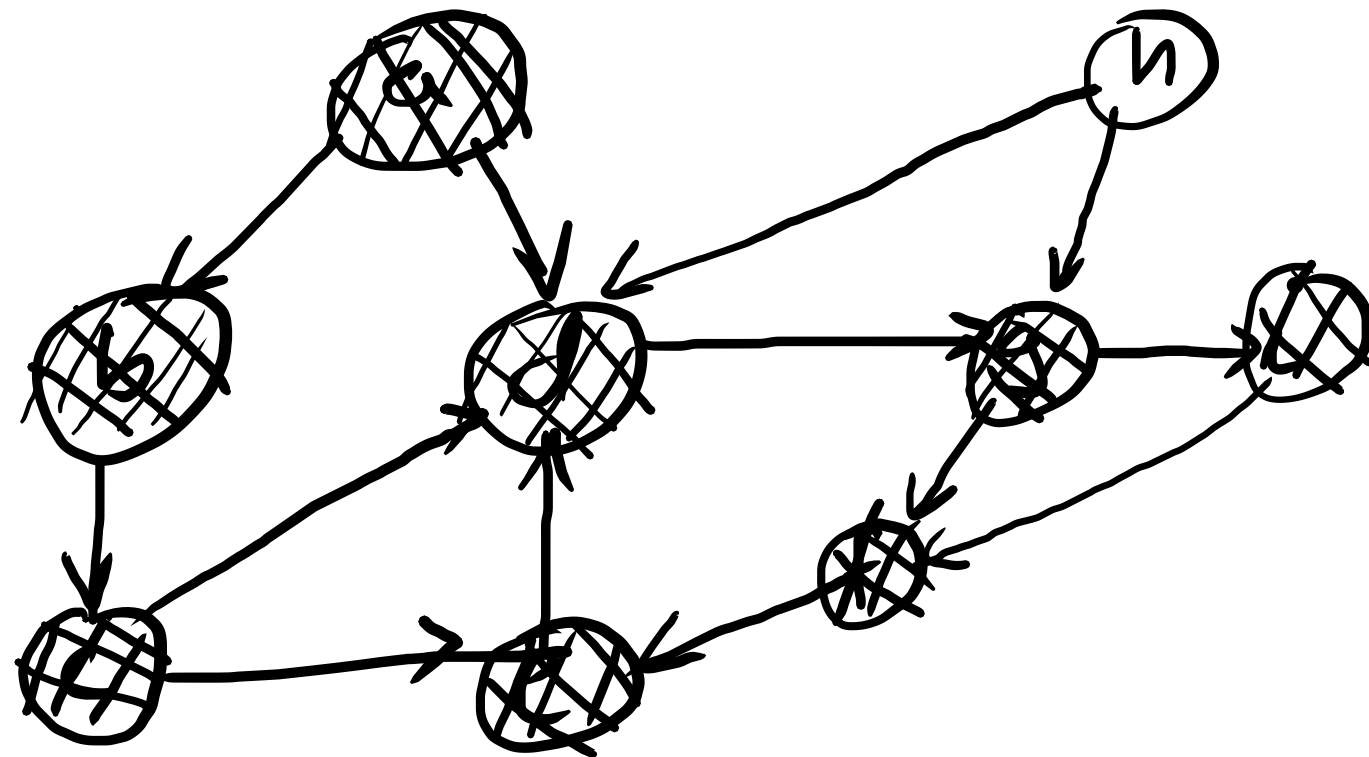
Dequeue v from Q

For every out-neighbor u of v that is still WHITE

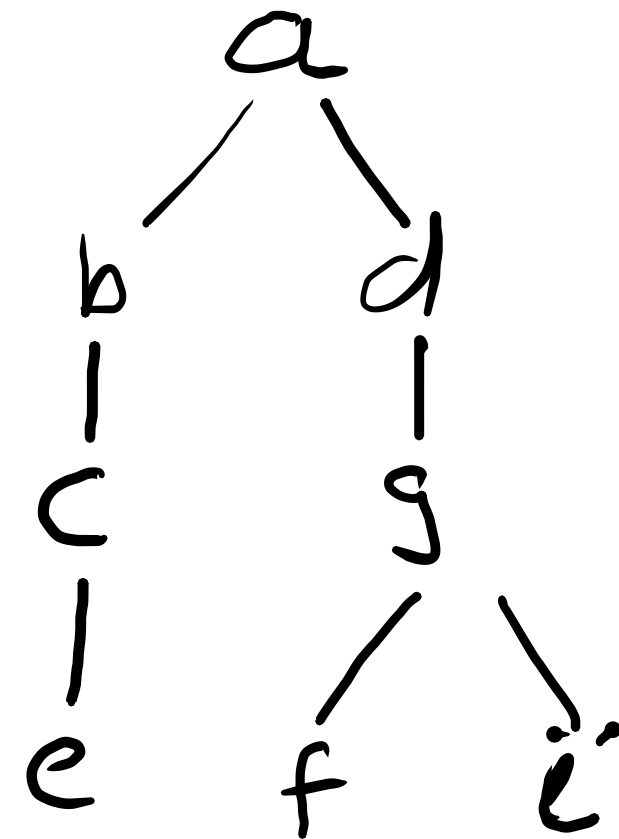
① Color u as GRAY

② Make u as child of v in Tree

Color u as BLACK



$$Q = \{ \}$$



BFS(G, s)

$G(V, E)$

for each vertex $u \in G.V - \{s\}$

$u.color = WHITE$

$u.d = \infty$

$u.\pi = Nil$

$s.color = GRAY$

$s.d = 0$

$s.\pi = Nil$

$Q = \emptyset$

ENQUEUE(Q, s)

while $Q \neq \emptyset$

$u = DEQ(Q)$

for each $v \in adj[u]$

if $v.color = WHITE$

$v.color = GRAY$

$v.d = u.d + 1$

$v.\pi = u$

ENQ(Q, v)

$u.color = BLACK$

A Source

